1.3.5 ▶ ▶ ▶ Total Kinetic Energy of Any System

This section a derivation for a frequently used form of the total kinetic energy of a system by separating the kinetic energy of the system into energy due to the center of mass and energy due to relative motion. Note that the section title indicates that Physics C students should review this concept, which I deem true because the topics here involve concepts used in rotational kinetic energy and the parallel-axis theorem. However, one could very well pass Physics C without the extra nuance provided by this proof[6].

First, consider an arbitrary system (rigid or non-rigid) with total mass M in a coordinate system with center of mass at $\mathbf{r}_{cm}$. Further, consider the ith at the position $\mathbf{r}_i$ and at a position $\mathbf{r}_{icm}$ relative to the center of mass.

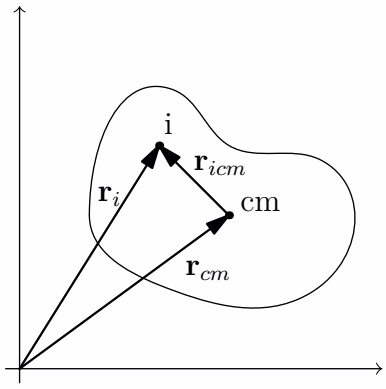

Fig. 1.6

Notice that $\mathbf{r}_i = \mathbf{r}_{cm} + \mathbf{r}_{icm}$ and thus $\mathbf{v}_i = \mathbf{v}_{cm} + \mathbf{v}_{icm}$. The kinetic energy of this particle is thus

$$\begin{aligned} K_i &= \frac{1}{2} m_i \mathbf{v}_i^2 \\ &= \frac{1}{2} m_i (\mathbf{v}_{cm} + \mathbf{v}_{icm})^2 \\ &= \frac{1}{2} m_i (v_{cm}^2 + 2\mathbf{v}_{cm} \cdot \mathbf{v}_{icm} + v_{icm}^2) \end{aligned} \tag{1.51}$$

Summing over all particles yields the following sum:

$$\begin{aligned} K_{total} = &\sum_i K_i = \\ &\sum_i \frac{1}{2} m_i v_{cm}^2 \\ &+ \sum_i m_i (\mathbf{v}_{cm} \cdot \mathbf{v}_{icm}) \\ &+ \sum_i \frac{1}{2} m_i v_{icm}^2 \end{aligned}$$

[6] F=ma and USAPhO students, though, should stay!

The first term is simply equal to $\frac{1}{2}Mv_{cm}^2$ because v_{cm} is independent of the ith particle. The second term is a bit more difficult to understand, though. Thus, prepare yourself! The next part gets quite dense. Rearranging the second term in a very specific way yields:

$$\mathbf{v}_{cm} \cdot \frac{d}{dt}\left(\sum_i m_i \mathbf{r}_{icm}\right) \tag{1.52}$$

To understand this new sum recall the definition of center of mass, namely that

$$M\mathbf{r}_{cm} = \sum_i m_i \mathbf{r}_i$$

This equation says that the product of the object's total mass M times the distance from the origin to the center of mass of a system $\mathbf{r}_{cm}$ is equal to the sum of the products of the mass of each constituent particle m_i and its distance to the origin $\mathbf{r}_i$. In eq. 1.51, however, notice that the distance in the sum $\mathbf{r}_{icm}$ is the distance of each particle from the center of mass, implying that the origin is centered on the center of mass. That means that the sum in eq. 1.51 is just equal to the total mass of the object M times the distance from the center of mass to the center of mass, which is zero!

Hence, we have

$$K_{total} = \frac{1}{2}Mv_{cm}^2 + \sum_i \frac{1}{2}m_i v_{icm}^2$$

Where v_{icm} is the relative velocity of the ith particle to the center of mass ★